# 郭　亨

guoheng@bupt.edu.cn | Google Scholar

个人主页

## 工作经历

北京邮电大学 — 北京, 中国
拔尖人才教授 — 2023 年 11 月 – 今

大阪大学 — 大阪, 日本
特任助理教授 — 2022 年 4 月 – 2023.10

## 教育背景

大阪大学 — 大阪, 日本
多媒体信息处理，博士 | 导师：*Yasuyuki Matsushita*，施柏鑫 — 2018 年 10 月 – 2022 年 3 月

电子科技大学 — 成都, 中国
信号与信息处理，硕士 | 导师：曾兵，刘帅成 — 2015 年 9 月 – 2018 年 7 月

电子科技大学 — 成都, 中国
电子信息工程，学士 — 2011 年 9 月 – 2015 年 7 月

## 主持项目

国家自然科学基金优秀青年科学基金项目（海外），**主持** — 2024 - 2026
国家自然科学基金面上项目，**主持** — 2024 - 2027
国家自然科学基金企业创新发展联合基金项目，**参与** — 2024 - 2027
京津冀基础研究专项，**主持** — 2024 - 2027
河北省重大科技支撑计划项目，**主持** — 2024 - 2026
北京市科技计划国际（港澳台）联合研发项目，**主持** — 2025 - 2027

## 代表性工作

(∗ Equal contribution, † Corresponding author.)

[1] Ruiyang Chen, Feiran Li, **Heng Guo**†, Zhanyu Ma. "FlashNormal: Detailed Surface Normal Estimation from Flash and No-Flash Images." IEEE Transactions on Circuits and Systems for Video Technology. (**TCSVT 2026**).

[2] Zonglin Li, Kexian Xie, Feiran Li, Chu Zhou, Yixing Huang, Ruiyang Chen, Zhanyu Ma, **Heng Guo**†. "AD-GS: Anatomical Density-guided 3D Gaussian Splatting for Sparse-View CBCT Reconstruction." British Machine Vision Conference. (**BMVC 2026**).

[3] Tongyu Yang, **Heng Guo**, Xingyou Lin, Yasuyuki Matsushita, Xin Fan. "Uncertainty-guided Neural Radiance Fields against Diverse Degradation Factors." (**Neurocomputing 2026**).

[4] Tengjiao Yin, Jinglei Shi, **Heng Guo**, Xi Wang. "VIGOR: Video Geometry-Oriented Reward for Temporal Generative Alignment." European Conference on Computer Vision. (**ECCV 2026**).

[5] Jilong Zhang, Songyun Yang, Yufei Han, **Heng Guo**†, Zhanyu Ma, "Photometric Stereo Prior Boosted Sparse Multiview Stereo." IEEE International Conference on Image Processing. (**ICIP 2026**).

[6] Tongyu Yang, **Heng Guo**, Yasuyuki Matsushita, Fumio Okura, Yu Luo, Xin Fan. "High-fidelity Multi-view Normal Integration with Scale-encoded Neural Surface Representation." Computer Graphics Forum. (**CGF 2026**).

[7] Zaiyan Yang, Xinpeng Liu, **Heng Guo**†, Jinglei Shi, Zhanyu Ma, Fumio Okura. "NRGS: Neural Regularization for Robust 3D Semantic Gaussian Splatting." IEEE International Conference on Multimedia and Expo. (**ICME 2026**).

[8] Zonglin Li, Jieji Ren, Shuangfan Zhou, **Heng Guo**†, Jinnuo Zhang, Jiang Zhou, Boxin Shi, Zhanyu Ma, Guoying Gu. "Near-Light Color Photometric Stereo for Mono-Chromaticity Non-Lambertian Surface." IEEE International Conference on Acoustics, Speech and Signal Processing. (**ICASSP 2026**).

[9] Jinyi Chang, Dongliang Chang, Lei Chen, Bingyao Yu, **Heng Guo**, Zhanyu Ma. "Towards Privacy-Preserving Fine-Grained Visual Classification via Hierarchical Learning from Label Proportions." IEEE International Conference on Acoustics, Speech and Signal Processing. (**ICASSP 2026**).

[10] Wenjie Li, **Heng Guo**†, Yuefeng Hou, Zhanyu Ma. "FourierSR: A Fourier Token-based Plugin for Efficient Image Super-Resolution." IEEE Transactions on Image Processing. (**TIP 2026**).

[11] Wenjie Li, Jinglei Shi, Jin Han, **Heng Guo**†, Zhanyu Ma. "Seeing Through the Rain: Resolving High-Frequency Conflicts in Deraining and Super-Resolution via Diffusion Guidance." AAAI Conference on Artificial Intelligence. (**AAAI 2026**).

[12] Zhenyu Jin, Wenjie Li, Zhanyu Ma, **Heng Guo**†. "SpecGen: Neural Spectral BRDF Generation via Spectral-Spatial Tri-plane Aggregation." IEEE Winter Conference on Applications of Computer Vision. (**WACV 2026**).

[13] Wenjie Li, Xiangyi Wang, **Heng Guo**†, Guangwei Gao, Zhanyu Ma. "Self-Supervised Selective-Guided Diffusion Model for Old-Photo Face Restoration." Annual Conference on Neural Information Processing Systems. (**NeurIPS 2025**).

[14] Wenjie Li, **Heng Guo**†, Yuefeng Hou, Guangwei Gao, Zhanyu Ma. "Dual-domain Modulation Network for Lightweight Image Super-Resolution." IEEE Transactions on Multimedia. (**TMM 2025**).

[15] Yufei Han, Bowen Tie, **Heng Guo**†, Youwei Lyu, Si Li†, Boxin Shi, Yunpeng Jia, Zhanyu Ma. "PolGS: Polarimetric Gaussian Splatting for Fast Reflective Surface Reconstruction." IEEE International Conference on Computer Vision. (**ICCV 2025**).

[16] Kailong Zhang, Youwei Lyu, **Heng Guo**†, Si Li, Zhanyu Ma, Boxin Shi. "PolarAnything: Diffusion-based Polarimetric Image Synthesis." IEEE International Conference on Computer Vision. (**ICCV 2025 Highlight**).

[17] Chu Zhou, Yixin Yang, Junda Liao, **Heng Guo**, Boxin Shi, Imari Sato. "Polarimetric Neural Field with Unified Complex-Valued Wavefunction." IEEE International Conference on Computer Vision. (**ICCV 2025**).

[18] Mingzhi Pei, Xu Cao, Xiangyi Wang, **Heng Guo**†, Zhanyu Ma. "PMNI: Pose-free Multi-view Normal Integration for Reflective and Textureless Surface Reconstruction." IEEE Conference on Computer Vision and Pattern Recognition. (**CVPR 2025**).

[19] Shuangfan Zhou, Chu Zhou, Youwei Lyu, **Heng Guo**†, Zhanyu Ma, Boxin Shi, Imari Sato. "PIDSR: Complementary Polarized Image Demosaicing and Super-Resolution." IEEE Conference on Computer Vision and Pattern Recognition. (**CVPR 2025**).

[20] Mengqiu Xu, Kaixin Chen, **Heng Guo**, Yixiang Huang, Ming Wu, Zhenwei Shi, Chuang Zhang, Jun Guo. "MFogHub: Bridging Multi-Regional and Multi-Satellite Data for Global Marine Fog Detection and Forecasting." IEEE Conference on Computer Vision and Pattern Recognition. (**CVPR 2025**).

[21] Songyun Yang, Yufei Han, Jilong Zhang, Kongming Liang, Peng Yu, Zhaowei Qu, **Heng Guo**. "RotatedMVPS: Multi-view Photometric Stereo with Rotated Natural Light." IEEE Conference on Multimedia and Expo. (**ICME 2025 Oral**).

[22] Youwei Lyu*, **Heng Guo***, Kailong Zhang, Si Li, Boxin Shi. "SfPUEL: Shape from Polarization under Unknown Environment." Annual Conference on Neural Information Processing Systems. (**NeurIPS 2024**).

[23] Li Wenjie, **Heng Guo**†, Xuannan Liu, Kongming Liang, Jiani Hu, Zhanyu Ma, Jun Guo. "Efficient Face Super-Resolution via Wavelet-Based Feature Enhancement Network." ACM International Conference on Multimedia. (**ACM MM 2024**).

[24] **Heng Guo***, Jieji Ren*, Feishi Wang*, Boxin Shi, Mingjun Ren, Yasuyuki Matsushita. "DiLiGenRT: A Photometric Stereo Dataset with Quantified Roughness and Translucency." IEEE Conference on Computer Vision and Pattern Recognition. (**CVPR 2024**).

[25] Yufei Han, **Heng Guo**†, Koki Fukai, Hiroaki Santo, Boxin Shi, Fumio Okura, Zhanyu Ma, Yunpeng Jia. "NeRSP: Neural 3D Reconstruction for Reflective Objects with Sparse Polarized Images." IEEE Conference on Computer Vision and Pattern Recognition. (**CVPR 2024**).

[26] Feishi Wang*, Jieji Ren*, **Heng Guo***, Mingjun Ren, Boxin Shi. "DiLiGenT-Pi: Photometric Stereo for Planar Surfaces with Rich Details –Benchmark Dataset and Beyond." International Conference on Computer Vision. (**ICCV 2023**).

[27] Jun Hoong Chan, Bohan Yu, **Heng Guo**, Jieji Ren, Zongqing Lu, Boxin Shi. "ReLeaPS: Reinforcement Learning-based Illumination Planning for Generalized Photometric Stereo." International Conference on Computer Vision. (**ICCV 2023**).

[28] Jipeng Lv, **Heng Guo**†, Guanying Chen, Jinxiu Liang, Boxin Shi. "Non-Lambertian Multispectral Photometric Stereo via Spectral Reflectance Decomposition." International Joint Conferences on Artificial Intelligence (**IJCAI 2023**).

[29] **Heng Guo**, Boxin Shi, Yasuyuki Matsushita. "Neural BRDF Plugin for Unsupervised Photometric Stereo." IEEE International Conference on Network Intelligence and Digital Content (**IC-NIDC 2023 best paper**).

[30] Feiran Li*, **Heng Guo***, Hiroaki Santo, Fumio Okura, Yasuyuki Matsushita. "Learning to Synthesize Photorealistic Dual-pixel Images from RGBD frames." International Conference on Computational Photography (**ICCP 2023**).

[31] Lilika Makabe, **Heng Guo**, Hiroaki Santo, Fumio Okura, Yasuyuki Matsushita. "Near-light Photometric Stereo with Symmetric Lights." International Conference on Computational Photography. (**ICCP 2023**).

[32] **Heng Guo**, Fumio Okura, Boxin Shi, Takuya Funatomi, Yasuhiro Mukaigawa, Yasuyuki Matsushita. "Multispectral Photometric Stereo for Spatially-Varying Spectral Reflectances." International Journal of Computer Vision. (**IJCV 2022**).

[33] **Heng Guo***, Zhipeng Mo*, Boxin Shi, Feng Lu, Sai-Kit Yeung, Ping Tan, Yasuyuki Matsushita. "Patch-based Uncalibrated Photometric Stereo under Natural Illumination." IEEE Transactions on Pattern Analysis and Machine Intelligence. (**TPAMI 2021**).

[34] **Heng Guo**, Fumio Okura, Boxin Shi, Takuya Funatomi, Yasuhiro Mukaigawa, Yasuyuki Matsushita. "Multispectral Photometric Stereo for Spatially-Varying Spectral Reflectances: A Well-posed Problem?" IEEE Conference on Computer Vision and Pattern Recognition. (**CVPR 2021**).

[35] **Heng Guo**, Boxin Shi, Michael Waechter, Yasuyuki Matsushita. "Self-calibrating Near-light Photometric Stereo under Anisotropic Light Emission." Meeting on Image Recognition and Understanding (**MIRU 2020 outstanding paper**).

[36] **Heng Guo**, Shuaicheng Liu, Shuyuan Zhu, Heng Tao Shen, Bing Zeng. "View-Consistent MeshFlow for Stereoscopic Video Stabilization." IEEE Transactions on Computational Imaging. (**TCI 2018**).

[37] **Heng Guo**, Shuaicheng Liu, Tong He, Shuyuan Zhu, Bing Zeng, Moncef Gabbouj. "Joint Video Stitching and Stabilization from Moving Cameras." IEEE Transactions on Image Processing (**TIP 2016**).

[38] **Heng Guo**, Shuaicheng Liu, Shuyuan Zhu, Bing Zeng. "Joint Bundled Camera Paths for Stereoscopic Video Stabilization." International Conference on Image Processing. (**ICIP 2016**) (Oral).

## 获奖荣誉

| | |
|---|---|
| 第六届北京高校教师教学创新大赛一等奖 | 2026 年 4 月 |
| 第六届全国高校教师教学创新大赛三等奖 | 2026 年 7 月 |
| International Conference on Network Infrastructure and Digital Content 最佳论文 (Top 1%) | 2023 年 10 月 |
| Meeting on Image Recognition and Understanding 最佳学生论文 (Top 1%) | 2020 年 4 月 |
| 北京市教育教学成果奖一等奖《面向大一新生的人工智能通识教育改革与创新》 | 2026 年 5 月 |
| 首届北京市大学生"人工智能 +"创新大赛二等奖《遗珍 · 神笔立像》(指导教师) | 2026 年 1 月 |
| 2025 年北京高校"优质本科课件"《人工智能导论》 | 2026 年 3 月 |
| 北京邮电大学优秀学士学位论文指导老师 | 2026 年 6 月 |

## 学术服务

- 中国电子学会虚拟现实分委会委员
- 中国计算机协会 CCF-YOCSEF 北京分委会委员
- Journal of the Optical Society of America (JOSA) 客座编辑
- 期刊审稿人: IEEE TPAMI, IJCV
- 会议审稿人: CVPR, ECCV, ICCV, NeurIPS, AAAI
- Demo Chair: ACCV 2026
- 论坛主席：第 24 期 CCF-CV 视界无限系列研讨会
- 2025 CSIG 青年科学家会议智能科学成像论坛召集人
- 2025 CAA 人工智能与机器人教育大会组织主席
- 论坛报告人：第 22 届日本信息科学与技术论坛 (情報科学技術フォーラム)